

# Desk Organizer Project Workbook

Write name here...

# Stage 1: Identify the problem

**Answer the following questions in complete sentences.**

What problem are you trying to solve with this desk organizer?
Example: “My pens and pencils roll around my desk, so I need a holder for them.”

Type here...

Who will use this organizer? What are their characteristics?
Example: I will be using this organizer. I am a Grade 10 student who carries...

Type here...

# Stage 1: Identify the problem

Who will use this organizer? Just you? Someone else too?

Type here...

Where will the organizer be used? Home, school, office?

Type here...

What are **important rules (constraints)** you must follow?
Example: size limit, safety, printer size, cost, material strength.

1. My organizer must...
2. My organizer must...
3. My organizer must...
4. My organizer must...
5. My organizer must...

# Stage 2: Plan and research

Find Examples: Look at three desk organizers online or in a store.
**Write one thing you like and one thing you don't like about each.**

| EXAMPLE | WHAT YOU LIKE | WHAT YOU DON'T LIKE |
|---|---|---|
| Put link here... | Type here... | Type here... |
| Put link here... | Type here... | Type here... |
| Put link here... | Type here... | Type here... |

# Stage 2: Plan and research

What material will you use for 3D printing? Why is it a good choice?

Type here...

Which 3D design software will you use? What are some tools and functions you will need to know how to use to build your model?

Type here...

# Stage 2: Plan and research

Write a simple timeline of steps you need to take.
Example: October 3, Build 3D model, Use TinkerCAD to create shapes...

| DATE | STEP | DESCRIPTION |
|---|---|---|
| Type here... | Type here... | Type here... |
| Type here... | Type here... | Type here... |
| Type here... | Type here... | Type here... |
| Type here... | Type here... | Type here... |
| Type here... | Type here... | Type here... |
| Type here... | Type here... | Type here... |

# Stage 3: Create ideas

Draw three different ideas for your organizer. Use labels to show parts.
Example: “Phone holder,” “Pen section,” “Paper clip area.”

# Stage 3: Create ideas

For each idea, write one strength (good thing) and one weakness (problem):

| IDEA | STRENGTH | WEAKNESS |
|---|---|---|
| 1 | Type here... | Type here... |
| 2 | Type here... | Type here... |
| 3 | Type here... | Type here... |

# Stage 4: Choose and model

Which idea did you choose? Why did you choose this idea?
Hint: talk about what makes it better than other designs, refer to strengths/weaknesses

Type here...

# Stage 4: Choose and model

Make a detailed **orthographic** drawing with measurements (length, width, height in millimeters), labels for each section, and space for the objects you will store.
Hint: be sure to include three views (top, front, and side)

# Stage 4: Choose and model

Use TinkerCAD to make a 3D model of your organizer. When finished, paste screenshots of your model here (from different angles).

# Stage 5: Build and test

Add a photo of your final 3D printed model here.

If you had any problems, add notes below to explain what they were.

Problems...

# Stage 5: Build and test

Use the following questions to test your design. Give each question a score from 1-10 and then explain why you gave it that score.

| TEST | SCORE | RATIONALE |
|---|---|---|
| Do your objects fit? | ?/10 | Type here... |
| Is it stable and strong? | ?/10 | Type here... |
| Does it meet the rules from Stage 1? | ?/10 | Type here... |

# Stage 6: Improve and reflect

**Answer the following questions in complete sentences.**

| | |
|---|---|
| What changes would you make to make your design better? | Type here... |
| What was the hardest part of the project? Why? | Type here... |
| What did you learn about 3D design and printing? | Type here... |
| If you were to design this for a company to sell, what would you change? Why? | Type here... |

# Stage 6: Improve and reflect

Use this checklist before handing in your project:

| | |
|---|---|
| **Stage 1:** Problem identified and five rules written | ✅ |
| **Stage 2:** Research completed and timeline made | |
| **Stage 3:** Three sketches with strengths and weaknesses | |
| **Stage 4:** Detailed orthographic drawing and 3D model created | |
| **Stage 5:** Printed and tested | |
| **Stage 6:** Reflection completed | |